

[Public](#)[Plan Vivo Nature Resources](#)[SOP: Vegetation Plot Registration](#)

Standard Operating Procedures - Vegetation Plot Registration

KEY DETAILS

Date : 11 May 2026

Version : 2026.1

Data type : Direct observation

Target groups : Woody Vegetation and Herbaceous Vegetation

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1. PREAMBLE

As a Data Analytics Provider for the Plan Vivo Nature biodiversity credit framework (PVN), Natural State is committed to the highest standards of scientific transparency and integrity. Data collection and processing methods for PVN projects are detailed in a set of Standard Operating Procedures (SOPs). These documents describe the steps involved in all project monitoring methodologies, from data collection to data processing and storage. Each SOP documents key methodological details for a specific data type and target group in accordance with the [PVN Methodology](#).

2. GLOSSARY

Additional definitions are available from Plan Vivo on the [PVN Documents page](#).

Belt transect : A type of elongated vegetation plot (see definition below) that is geographically delineated by a midpoint, two endpoints, and a lateral width. For the purposes of this SOP, a belt transect is specified as 50 m long and 5 m wide.

DAP : Data Analytics Provider. Defined by Plan Vivo as "An independent organisation approved by Plan Vivo to conduct the services of data analysis for the calculation of Plan Vivo Biodiversity Certificates (PVBCs) for Projects in line with the PV Nature Methodology and requirements."

Dominant woody species : The primary species that provides structure to the habitat within or around a sampling unit. Must stand taller than the height of the grass. The dominant woody vegetation might not be the most abundant type but is rather the one that consistently provides maximum vertical structure within the plot or area of interest.

Geohash : A geographic multi-scale encoding system that discretises the Earth's surface into a hierarchy of rectangular cells, each identified by a unique alphanumeric code. Natural State uses geohashes to define vegetation plot locations, with each plot midpoint corresponding to the centroid of a geohash cell.

ODK : ODK (Open Data Kit) is a free, open-source platform that lets organizations build custom digital forms for collecting data in the field — even without an internet connection. It's widely used by researchers, nonprofits, and government agencies to gather structured data. Natural State uses ODK for all field data collection for PVN projects where Natural State serves as the Data Analytics Provider.

Pillar Metric : Defined by Plan Vivo as "A metric that tracks change in one specific aspect of site-level Biodiversity. The Pillar Metrics are aggregated into the Multimetric to produce a single value representing site-level biodiversity change." See the PVN methodology for further details.

PVN : Plan Vivo Nature biodiversity framework. A standard for accrediting and certifying projects that restore or conserve biodiversity in accordance with Plan Vivo's core values of community engagement and fair and equitable benefit sharing.

PVN Methodology : Defined by Plan Vivo as "The biodiversity quantification methodology for PV Nature, by which Plan Vivo Biodiversity Certificates (PVBCs) are quantified and quality checked. It

sets minimum standards on Biodiversity data quality and sampling protocols, and it defines the Pillar Metrics via which Biodiversity data is aggregated into a Multimetric and then converted into numbers of PVBCs. Under the methodology, projects must collect and report data on a number of broad Target Groups, across a range of trophic levels."

Target Group : Defined by Plan Vivo as "A broad assemblage of species that will be surveyed using the same Data Collection Tool. This is for example, 'fish' rather than a specific type of fish."

Vegetation plot : A two-dimensional defined area within which vegetation biodiversity is sampled either across the entirety of the area or following a systematic sub-sampling spatial design. Every vegetation plot includes a midpoint that corresponds to the centroid of a geohash. For the purposes of this SOP, the plot is defined as a belt transect (see definition above) that is 50 m long and 5 m wide.

3. METHODS OVERVIEW

This SOP details procedures for registering a 50 m by 5 m belt transect vegetation plot in ODK for sampling as part of the requisite monitoring of a PVN project. Vegetation plots delineate areas within which monitoring teams make direct observations of plant species to support calculations of Pillars I-III of the PVN Methodology. Registering vegetation plots is a prerequisite for collecting data on all plant target groups. In addition to georeferencing the plot area, the plot registration also captures plot metadata that provide essential context for species identification and independent verification. Figure 1 shows the layout of a standard vegetation plot, illustrating the positions of the midpoint, endpoints A and B, and the 5 m belt width relative to the transect midline.

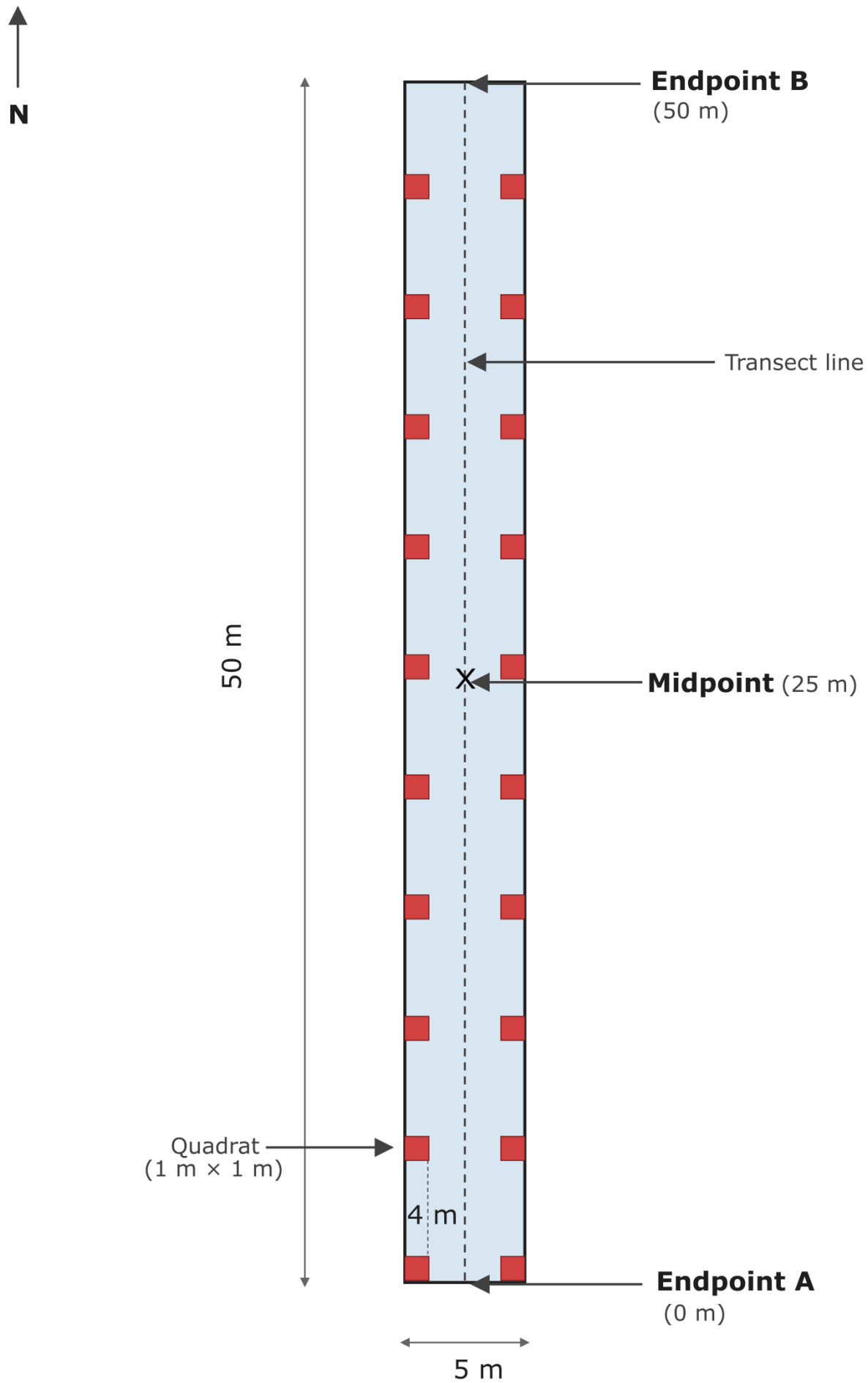


Figure 1. Schematic of a 50 m × 5 m belt transect vegetation plot. The plot is defined by a midpoint and two endpoints (A and B), each 25 m away from the midpoint. The 5 m width (2.5 m on each side of the tape measure) is measured perpendicular to the transect midline.

3.1 METHODOLOGICAL BACKGROUND

The procedures described here are intended for any vegetation plots defined using a belt transect and are not tailored to any specific survey protocol. Rather, the plot registration procedures are intended to provide a flexible way to georeference plots and to capture plot characteristics, including photographic records. The protocol is implemented through the [Register, Vegetation Plot](#) ODK form.

4. SAMPLING PREPARATION

- The equipment mentioned in the list below needs to be gathered, checked and packed before sampling begins.
- Handheld GPS devices need to be at full battery power and need to be preloaded with the `baseline_sampling_points.gpx` file.
- Mobile devices should be double checked to ensure they are at full battery power and that all necessary ODK forms are loaded and refreshed.

4.1 EQUIPMENT LIST

- A GPS-enabled Android mobile device or tablet outfitted with a working camera and preloaded with ODK and the relevant project forms.
- A handheld GPS pre-loaded with the survey site point locations. A GPS app on a mobile device can be used. The ODK App should not be used for navigating to sites.
- A tape measure at least 50 m long.
- A compass.
- A 2.5 m measuring stick or a 3 m or 5 m tape measure.
- Three stakes or other markers to delineate the midpoint and two endpoints of the belt transect (recommended).

- Extra batteries or charging banks for all electronic devices.

5. SAMPLING PROCEDURES

1. Navigate to the prescribed sampling location using the handheld GPS. This point represents the midpoint of the belt transect.
2. Open the ODK veg plot registration form and fill in the field team details and survey details sections.
3. Select the plot in the ODK form. Survey the surrounding area and take note of any reasons why sampling at this location is not viable either for safety reasons or because the site cannot provide valid data. Recording a site as non-viable is at the discretion of the field team but it must be clearly justified. Reasons other than safety why a site might be deemed non-viable include an obvious mismatch between the assigned stratum and actual habitat type or severe, unrepresentative disturbance from causes other than humans such as fire. Plots that have been damaged by a fire and have not had sufficient time for vegetation to regrow should not be sampled. If the reason a site is not viable is temporary (e.g., presence of dangerous animals) do not mark the site as not viable and instead return to the site at a later time.
4. If using stakes or other markers, place one at the midpoint of the belt transect. Then use the compass and tape measure to lay out the transect along the north/south axis. Ensure the tape measure is taut and is not entangled in vegetation.
5. Walk the transect with the measuring stick perpendicular to the tape, and ensure that there are no obstacles or hazards within the belt width. If there are, move or rotate the transect the shortest possible distance to avoid the obstacle or hazard. Transects must initially be laid out along the north south axis to prevent bias, but can be rotated to avoid obstacles or hazards.
The midpoint cannot be moved more than 25 m from the prescribed location. If no plot area free of obstacles or hazards is available within 25 m of the prescribed location, the plot should be recorded as non-viable and replaced with a backup sampling location. Obstacles to be avoided include:
 - roads with non-natural substrates (most narrow tracks are fine as long as they are roughly perpendicular to the transect)
 - buildings or other human-made permanent structures
 - permanent or semi-permanent non-vegetated habitat features -- e.g., standing or running water, dry streambeds, large boulders, pans, mud or dust wallows -- that cover at least 5 sq. m of the belt transect.

6. When satisfied that the transect has no impediments, fill in the plot details section of the ODK form. Then navigate to endpoint A of the transect and take a photo and record your location in the form. It does not matter which endpoint is A and which is B.
7. Proceed to the midpoint and endpoint B of the transect and continue recording photos and georeferencing the transect as prescribed in the ODK form.
8. Review the form for any typos, especially in free text fields. When satisfied all information is correct, confirm the submission and enter any other contextual notes in the comment field.
9. Submit the form.
10. Update ODK forms to ensure that the newly registered plot entity is reflected in other survey forms.
11. Repeat steps 1:10 for all remaining sampling locations.

6. DATA UPLOADS

Ensure that all saved forms are submitted as soon as a stable internet connection is available.

7. OTHER RELEVANT SOPS

PVN Woody Vegetation SOP

PVN Herbaceous Vegetation SOP

8. REVISION AND VERSION HISTORY AND DESCRIPTION

v2026.1 The version of record for the initiation of baseline data collection for the Coexisting with Elephants project (May 2026).